SCHENKTRONICS

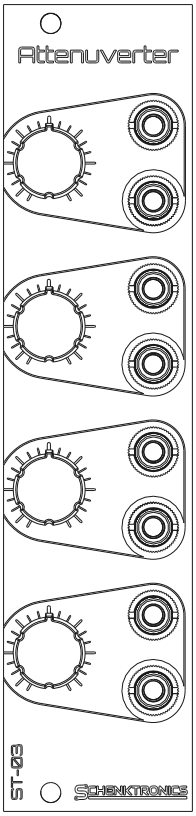
ST-03 Attenuverter

User Manual

For PCB v1.1 - Updated 2026-10-17

Official version: github.com/Schenktronics/ST03-attenuverter

Overview



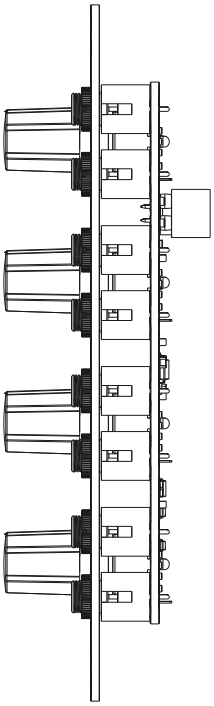
The Attenuverter is a 6HP module with four identical channels. Each channel scales the signal at its input by an amount set by its knob, anywhere from full level, through zero, to full level inverted.

Each input is normalled to +5 V. With nothing plugged into an input, its output is a steady voltage from -5 V to +5 V, set by the knob, so each channel doubles as a manual offset or CV source.

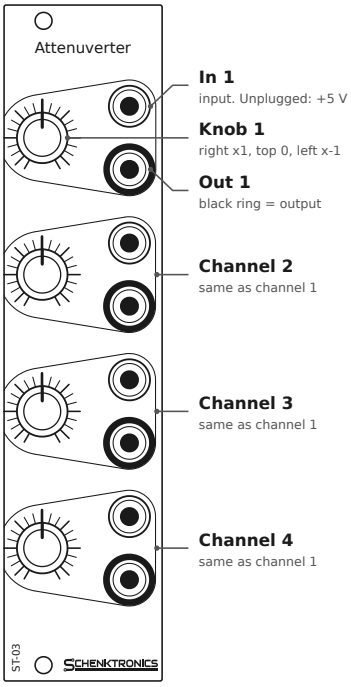
It's an active module, so it needs ±12 V from your case's power supply.

Specifications

Format	Eurorack, 6HP
Panel	30 × 128.5 mm
Depth	20 mm
Power	16 mA on +12 V, 9 mA on -12 V, no +5 V. 10-pin header.
Protection	Reverse-power diodes and resettable fuses on both rails, and clamp diodes on each input
Jacks	8 × 3.5 mm mono (TS): 4 inputs, 4 outputs
Controls	4 x knobs with a printed zero mark (100 kΩ linear potentiometers)
Range	x-1 (fully counterclockwise) to 0 (center) to x1 (fully clockwise)
Unplugged inputs	+5 V, so the output is -5 V to +5 V
Input impedance	About 34-53 kΩ, depending on the knob position: lowest fully counterclockwise (calculated)
Output impedance	1 kΩ
Signals	Audio, CV, gates, and bipolar signals. DC-coupled.
Output swing	About ±10 V, limited by the op-amp running on ±12 V



Panel layout



The panel has four identical channels, numbered 1 to 4 from the top. Each channel is outlined, with its knob on the left and two jacks on the right. The top jack is the input, and the jack with the **black ring** is the output.

Using the module

Knobs

Each knob sets how much of the input reaches the output, and which way up:

Knob position	Output
Fully clockwise	The input, unchanged (x1)
Halfway between the center and fully clockwise	About a third of the input (x0.36)
Center	Nothing (x0)
Halfway between the center and fully counterclockwise	About a third of the input, inverted (x-0.36)
Fully counterclockwise	The input, inverted (x-1)

"Inverted" means every voltage is negated: a +3 V CV becomes -3 V, and a rising envelope falls instead.

The panel has a long mark at the top of each knob's travel, where the output is zero. Line the knob's pointer up with it to turn a channel off.

The knobs are center-weighted. Halfway to either end gives about a third of the signal rather than half, so more of the knob's travel goes to small amounts, where fine control is hardest to get.

Normalled inputs: offset voltages

Each input is normalled to +5 V. When nothing is plugged into an input, the channel gets +5 V instead, and the output is a steady voltage:

- **Fully clockwise:** +5 V
- **Center:** 0 V
- **Fully counterclockwise:** -5 V

Use it as a manual CV knob for a module that doesn't have one, to transpose a sequence, or to shift an LFO up or down through a DC mixer. Plugging a cable into the input breaks the normal.

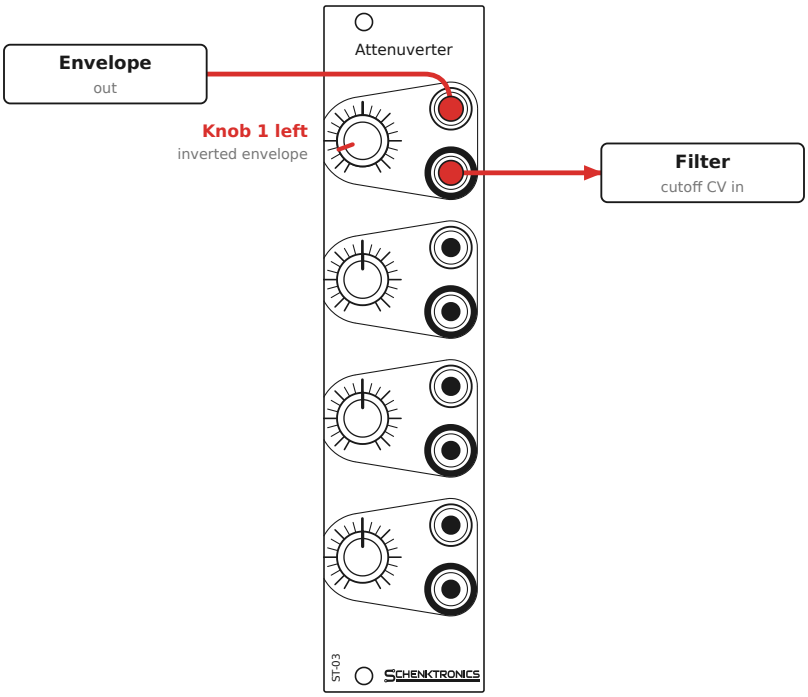
Using it on audio

On audio, a channel is a volume control that can also flip the phase. Mixing a signal with an inverted copy of itself cancels it out.

Patch examples

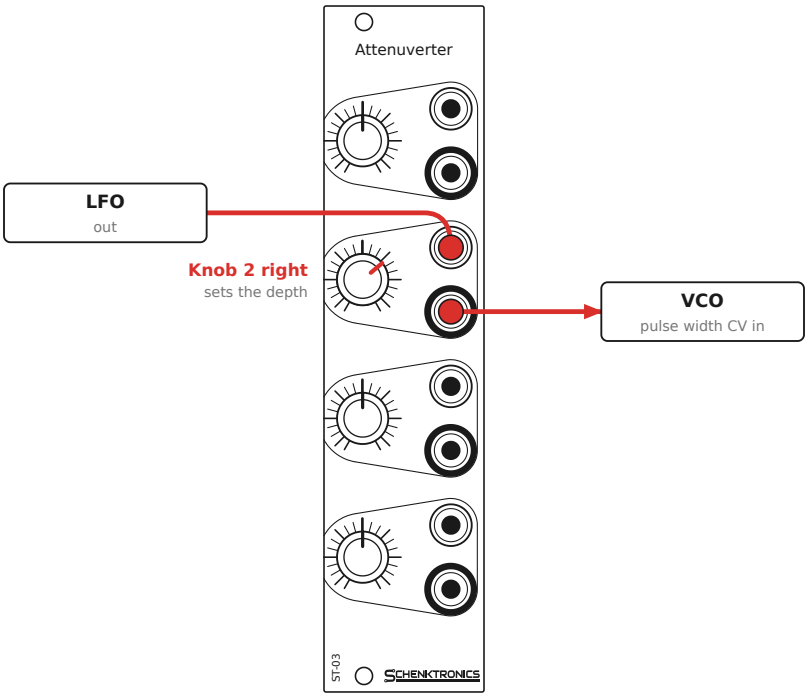
Each drawing shows the module sending the signal on the left and the modules receiving it on the right.

Invert an envelope



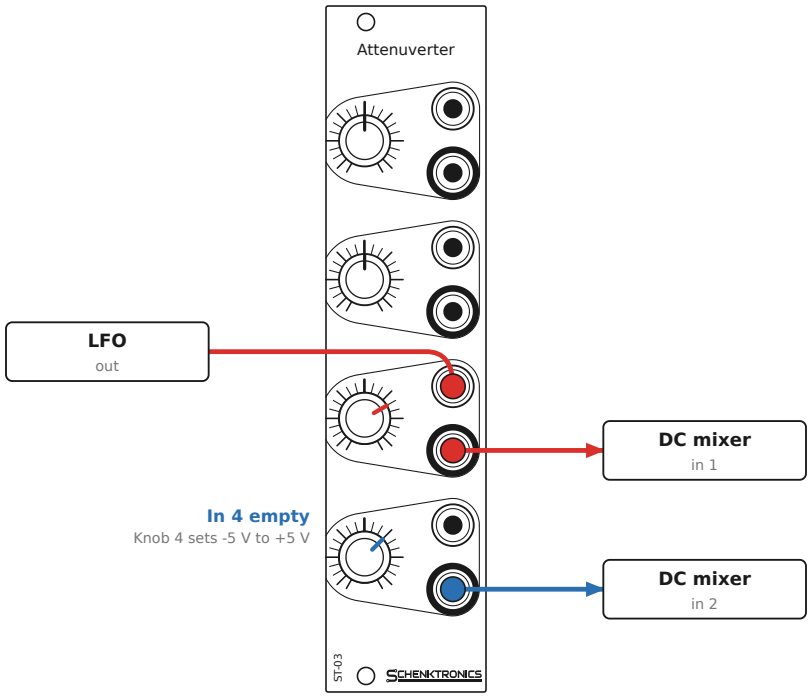
Turn Knob 1 left of center and the envelope comes out inverted: instead of opening the filter, each note closes it. Raise the filter's cutoff knob so the envelope has somewhere to sweep down from.

Set the depth of an LFO



Plenty of CV inputs have no level control, so a full-size LFO sweeps them from end to end. Patch the LFO through a channel and use its knob to set the depth. Just right of center gives a small movement, and left of center reverses the LFO's direction.

Offset an LFO



Channel 3 sets the LFO's depth. In 4 is empty, so Knob 4 gives a steady voltage between -5 V and +5 V. Add the two in a DC-coupled mixer, and the LFO moves around a center point you set with Knob 4.

Good practice

- **Don't patch outputs together.** Joining an Out jack to another module's output makes the two outputs fight, and the result is unpredictable. The 1 kΩ output resistors help protect this module, but the other module may not be protected. Use a mixer to combine signals.
- **An empty input isn't silent.** An unplugged input carries +5 V, so its output isn't 0 V unless the knob is at the center. If a patch has an unexpected offset, look for a channel with an empty input.
- **The zero mark is close to zero, not exact.** Pots vary, so there may be a very small signal left even with the pointer lined up. If you need true silence, unplug the output.
- **Pitch CV (V/oct).** Scaling a pitch CV changes the intervals, so the oscillator won't play in tune. Fully clockwise is close to x1, but it isn't calibrated for 1 V/oct. To transpose, use an empty input as an offset and add it to the pitch CV in a DC mixer.
- **Signals above about ±10 V clip.** The op-amp can't swing further than that on ±12 V. Standard Eurorack signals fit.

Installation

1. Turn your case off.
2. Connect a 10-pin to 16-pin Eurorack power cable to the header on the back of the module. The red stripe goes to **-12 V**, marked "Red Stripe" and "-12v" on the board. At the bus board end, the red stripe also goes to -12 V, which is usually marked on the bus.

3. Put the module in any free 6HP space.
4. Fix it in place with two M3 rack screws. Don't overtighten them.
5. Turn the case on, and check that turning a knob with an empty input moves its output between -5 V and +5 V.

The module is protected against a cable plugged in backwards, but don't rely on it. Always check the red stripe before you power on.

Circuit

Each channel is one section of a TL074 quad op-amp (U1), wired as a differential amplifier. Each input jack goes through a 1 kΩ resistor to a node. Two clamp diodes hold that node between +12 V and -12 V, so a signal beyond about ±12 V can't reach the op-amp. From the node, the signal goes two ways: through a 100 kΩ resistor to the op-amp's inverting input, and to one end of the knob's potentiometer, whose other end is grounded. The pot's wiper drives the op-amp's non-inverting input. With equal 100 kΩ resistors, the output is 2 x (wiper) - (input), so it runs from -1 times the input with the wiper at ground to +1 times the input with the wiper at the top, and is zero at the center. Two 100 kΩ resistors across the two halves of each pot give the center-weighted response. A 10 pF capacitor in the feedback loop keeps the op-amp stable, and a 1 kΩ resistor protects each output.

The inputs are switched jacks. Their normalling contacts connect to a 5.0 V reference (U2, an LM4040, fed from +12 V through a 1 kΩ resistor), so each unplugged input sees +5 V.

The power input has a resettable fuse and a reverse-biased diode on each rail. If the power cable is plugged in backwards, the diodes conduct and the fuses trip, protecting the circuit. The schematic is in hardware/attenuverter/Attenuverter.kicad_sch (https://github.com/Schenktronics/ST03-attenuverter/blob/main/hardware/attenuverter/Attenuverter.kicad_sch).